



```
In [ ]: '''
        Practical No. 01

        Title: of the Assignment: Data Wrangling, I

        Perform the following operations using Python on any open source dataset (e.g.
        Import all the required Python Libraries.
        1. Locate open source data from the web (e.g. https://www.kaggle.com).
        2. Provide a clear description of the data and its source (i.e., URL of the we
        3. Load the Dataset into the pandas data frame.
        4. Data Preprocessing: check for missing values in the data using pandas insul
        function to get some initial statistics. Provide variable descriptions. Types
        Check the dimensions of the data frame.
        5. Data Formatting and Data Normalization: Summarize the types of variables by
        the data types (i.e., character, numeric, integer, factor, and logical) of the
        data set. If variables are not in the correct data type, apply proper type cor
        6. Turn categorical variables into quantitative variables in Python.
        Objective :
        Students should be able to perform thedata wrangling operation using Python or
        '''
```

```
In [ ]: import pandas as pd
```

```
In [ ]: df=pd.read_csv("iris.csv")
```

```
In [ ]: df
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Speci
0	1	5.1	3.5	1.4	0.2	Ir setc
1	2	4.9	3.0	1.4	0.2	Ir setc
2	3	4.7	3.2	1.3	0.2	Ir setc
3	4	4.6	3.1	1.5	0.2	Ir setc
4	5	5.0	3.6	1.4	0.2	Ir setc
...	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	Ir virgini
146	147	6.3	2.5	5.0	1.9	Ir virgini
147	148	6.5	3.0	5.2	2.0	Ir virgini
148	149	6.2	3.4	5.4	2.3	Ir virgini
149	150	5.9	3.0	5.1	1.8	Ir virgini

150 rows × 6 columns

```
In [ ]: df.describe()
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthC
count	150.000000	150.000000	150.000000	150.000000	150.0000
mean	75.500000	5.843333	3.054000	3.758667	1.1986
std	43.445368	0.828066	0.433594	1.764420	0.7631
min	1.000000	4.300000	2.000000	1.000000	0.1000
25%	38.250000	5.100000	2.800000	1.600000	0.3000
50%	75.500000	5.800000	3.000000	4.350000	1.3000
75%	112.750000	6.400000	3.300000	5.100000	1.8000
max	150.000000	7.900000	4.400000	6.900000	2.5000

```
In [ ]: df.isnull()
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	False	False	False	False	False	F
1	False	False	False	False	False	F
2	False	False	False	False	False	F
3	False	False	False	False	False	F
4	False	False	False	False	False	F
...	...	...	...	...	...	
145	False	False	False	False	False	F
146	False	False	False	False	False	F
147	False	False	False	False	False	F
148	False	False	False	False	False	F
149	False	False	False	False	False	F

150 rows × 6 columns

```
In [ ]: df.isnull().sum()
```

```
Out[ ]: Id                0
SepalLengthCm          0
SepalWidthCm           0
PetalLengthCm          0
PetalWidthCm           0
Species                0
dtype: int64
```

```
In [ ]: df.duplicated()
```

```
Out[ ]: 0      False
1      False
2      False
3      False
4      False
...
145    False
146    False
147    False
148    False
149    False
Length: 150, dtype: bool
```

```
In [ ]: df.duplicated().sum()
```

```
Out[ ]: np.int64(0)
```

```
In [ ]: df.isnull().any()
```

```
Out[ ]: Id                False
SepalLengthCm           False
SepalWidthCm            False
PetalLengthCm           False
PetalWidthCm            False
Species                 False
dtype: bool
```

```
In [ ]: df.isna()
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	False	False	False	False	False	F
1	False	False	False	False	False	F
2	False	False	False	False	False	F
3	False	False	False	False	False	F
4	False	False	False	False	False	F
...	...	...	...	...	...	...
145	False	False	False	False	False	F
146	False	False	False	False	False	F
147	False	False	False	False	False	F
148	False	False	False	False	False	F
149	False	False	False	False	False	F

150 rows × 6 columns

```
In [ ]: df.head()
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [ ]: df.tail()
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Speci
145	146	6.7	3.0	5.2	2.3	Ir virgini
146	147	6.3	2.5	5.0	1.9	Ir virgini
147	148	6.5	3.0	5.2	2.0	Ir virgini
148	149	6.2	3.4	5.4	2.3	Ir virgini
149	150	5.9	3.0	5.1	1.8	Ir virgini

```
In [ ]: df.head(10)
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris- setosa
1	2	4.9	3.0	1.4	0.2	Iris- setosa
2	3	4.7	3.2	1.3	0.2	Iris- setosa
3	4	4.6	3.1	1.5	0.2	Iris- setosa
4	5	5.0	3.6	1.4	0.2	Iris- setosa
5	6	5.4	3.9	1.7	0.4	Iris- setosa
6	7	4.6	3.4	1.4	0.3	Iris- setosa
7	8	5.0	3.4	1.5	0.2	Iris- setosa
8	9	4.4	2.9	1.4	0.2	Iris- setosa
9	10	4.9	3.1	1.5	0.1	Iris- setosa

```
In [ ]: df.tail(10)
```

Out[]:		Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Speci
	140	141	6.7	3.1	5.6	2.4	Ir virgini
	141	142	6.9	3.1	5.1	2.3	Ir virgini
	142	143	5.8	2.7	5.1	1.9	Ir virgini
	143	144	6.8	3.2	5.9	2.3	Ir virgini
	144	145	6.7	3.3	5.7	2.5	Ir virgini
	145	146	6.7	3.0	5.2	2.3	Ir virgini
	146	147	6.3	2.5	5.0	1.9	Ir virgini
	147	148	6.5	3.0	5.2	2.0	Ir virgini
	148	149	6.2	3.4	5.4	2.3	Ir virgini
	149	150	5.9	3.0	5.1	1.8	Ir virgini

```
In [ ]: df.shape
```

```
Out[ ]: (150, 6)
```

```
In [ ]: df.index
```

```
Out[ ]: RangeIndex(start=0, stop=150, step=1)
```

```
In [ ]: df.tail(n=15)
```

Out[]:		Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Speci
	135	136	7.7	3.0	6.1	2.3	Ir virginia
	136	137	6.3	3.4	5.6	2.4	Ir virginia
	137	138	6.4	3.1	5.5	1.8	Ir virginia
	138	139	6.0	3.0	4.8	1.8	Ir virginia
	139	140	6.9	3.1	5.4	2.1	Ir virginia
	140	141	6.7	3.1	5.6	2.4	Ir virginia
	141	142	6.9	3.1	5.1	2.3	Ir virginia
	142	143	5.8	2.7	5.1	1.9	Ir virginia
	143	144	6.8	3.2	5.9	2.3	Ir virginia
	144	145	6.7	3.3	5.7	2.5	Ir virginia
	145	146	6.7	3.0	5.2	2.3	Ir virginia
	146	147	6.3	2.5	5.0	1.9	Ir virginia
	147	148	6.5	3.0	5.2	2.0	Ir virginia
	148	149	6.2	3.4	5.4	2.3	Ir virginia
	149	150	5.9	3.0	5.1	1.8	Ir virginia

```
In [ ]: df.columns
```

```
Out[ ]: Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
              'Species'],
              dtype='object')
```

```
In [ ]: df.dtypes
```

```
Out[ ]: Id                int64
SepalLengthCm          float64
SepalWidthCm           float64
PetalLengthCm          float64
PetalWidthCm           float64
Species                object
dtype: object
```

```
In [ ]: df.columns.values
```

```
Out[ ]: array(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm',
              'PetalWidthCm', 'Species'], dtype=object)
```

```
In [ ]: df.describe(include='all')
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
unique	NaN	NaN	NaN	NaN	NaN
top	NaN	NaN	NaN	NaN	NaN
freq	NaN	NaN	NaN	NaN	NaN
mean	75.500000	5.843333	3.054000	3.758667	1.198000
std	43.445368	0.828066	0.433594	1.764420	0.763000
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
In [ ]: df["SepalLengthCm"]
```

```
Out[ ]: 0      5.1
1      4.9
2      4.7
3      4.6
4      5.0
...
145    6.7
146    6.3
147    6.5
148    6.2
149    5.9
Name: SepalLengthCm, Length: 150, dtype: float64
```

```
In [ ]: df.sort_index(axis=1,
```



```
ascending=False)
```

Out[]:

	Species	SepalWidthCm	SepalLengthCm	PetalWidthCm	PetalLengthCm	
0	Iris-setosa	3.5	5.1	0.2	1.4	
1	Iris-setosa	3.0	4.9	0.2	1.4	
2	Iris-setosa	3.2	4.7	0.2	1.3	
3	Iris-setosa	3.1	4.6	0.2	1.5	
4	Iris-setosa	3.6	5.0	0.2	1.4	
...	...	...	...	...	...	
145	Iris-virginica	3.0	6.7	2.3	5.2	1
146	Iris-virginica	2.5	6.3	1.9	5.0	1
147	Iris-virginica	3.0	6.5	2.0	5.2	1
148	Iris-virginica	3.4	6.2	2.3	5.4	1
149	Iris-virginica	3.0	5.9	1.8	5.1	1

150 rows × 6 columns

```
In [ ]: df.sort_values(by="SepalLengthCm")
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Speci
13	14	4.3	3.0	1.1	0.1	Ir setc
8	9	4.4	2.9	1.4	0.2	Ir setc
42	43	4.4	3.2	1.3	0.2	Ir setc
38	39	4.4	3.0	1.3	0.2	Ir setc
41	42	4.5	2.3	1.3	0.3	Ir setc
...	...	...	...	...	...	
122	123	7.7	2.8	6.7	2.0	Ir virgini
117	118	7.7	3.8	6.7	2.2	Ir virgini
118	119	7.7	2.6	6.9	2.3	Ir virgini
135	136	7.7	3.0	6.1	2.3	Ir virgini
131	132	7.9	3.8	6.4	2.0	Ir virgini

150 rows × 6 columns

```
In [ ]: df.iloc[5]
```

```
Out[ ]: Id                6
SepalLengthCm          5.4
SepalWidthCm           3.9
PetalLengthCm           1.7
PetalWidthCm            0.4
Species                Iris-setosa
Name: 5, dtype: object
```

```
In [ ]: df[0:3]
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris- setosa
1	2	4.9	3.0	1.4	0.2	Iris- setosa
2	3	4.7	3.2	1.3	0.2	Iris- setosa

```
In [ ]: df.loc[:, ["PetalLengthCm", "PetalWidthCm"]]
```

```
Out[ ]:
```

	PetalLengthCm	PetalWidthCm
0	1.4	0.2
1	1.4	0.2
2	1.3	0.2
3	1.5	0.2
4	1.4	0.2
...	...	...
145	5.2	2.3
146	5.0	1.9
147	5.2	2.0
148	5.4	2.3
149	5.1	1.8

150 rows × 2 columns

```
In [ ]: df.iloc[:5, :]
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [ ]: df.iloc[:, :10]
```

Out[]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Speci
0	1	5.1	3.5	1.4	0.2	Ir setc
1	2	4.9	3.0	1.4	0.2	Ir setc
2	3	4.7	3.2	1.3	0.2	Ir setc
3	4	4.6	3.1	1.5	0.2	Ir setc
4	5	5.0	3.6	1.4	0.2	Ir setc
...	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	Ir virgini
146	147	6.3	2.5	5.0	1.9	Ir virgini
147	148	6.5	3.0	5.2	2.0	Ir virgini
148	149	6.2	3.4	5.4	2.3	Ir virgini
149	150	5.9	3.0	5.1	1.8	Ir virgini

150 rows × 6 columns

In []:

```
df.iloc[:10, :10]
```

Out[]:		Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
	0	1	5.1	3.5	1.4	0.2	Iris-setosa
	1	2	4.9	3.0	1.4	0.2	Iris-setosa
	2	3	4.7	3.2	1.3	0.2	Iris-setosa
	3	4	4.6	3.1	1.5	0.2	Iris-setosa
	4	5	5.0	3.6	1.4	0.2	Iris-setosa
	5	6	5.4	3.9	1.7	0.4	Iris-setosa
	6	7	4.6	3.4	1.4	0.3	Iris-setosa
	7	8	5.0	3.4	1.5	0.2	Iris-setosa
	8	9	4.4	2.9	1.4	0.2	Iris-setosa
	9	10	4.9	3.1	1.5	0.1	Iris-setosa

```
In [ ]: df.iloc[3:5, 0:2]
```

Out[]:		Id	SepalLengthCm
	3	4	4.6
	4	5	5.0

```
In [ ]: df.iloc[[1, 2,4], [0, 2]]
```

Out[]:		Id	SepalWidthCm
	1	2	3.0
	2	3	3.2
	4	5	3.6

```
In [ ]: df.iloc[1:3, :]
```

```
Out[ ]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa

```
In [ ]: df.iloc[:, 1:3]
```

```
Out[ ]:
```

	SepalLengthCm	SepalWidthCm
0	5.1	3.5
1	4.9	3.0
2	4.7	3.2
3	4.6	3.1
4	5.0	3.6
...	...	...
145	6.7	3.0
146	6.3	2.5
147	6.5	3.0
148	6.2	3.4
149	5.9	3.0

150 rows × 2 columns

```
In [ ]: df.iloc[1, 1]
```

```
Out[ ]: np.float64(4.9)
```

```
In [ ]: df['SepalLengthCm'].iloc[5]
```

```
Out[ ]: np.float64(5.4)
```

```
In [ ]: cols_2_4=df.columns[2:4]
df[cols_2_4]
```

```
Out[ ]:
```

	SepalWidthCm	PetalLengthCm
0	3.5	1.4
1	3.0	1.4
2	3.2	1.3
3	3.1	1.5
4	3.6	1.4
...	...	...
145	3.0	5.2
146	2.5	5.0
147	3.0	5.2
148	3.4	5.4
149	3.0	5.1

150 rows × 2 columns

```
In [ ]: df[df.columns[2:4]].iloc[5:10]
```

```
Out[ ]:
```

	SepalWidthCm	PetalLengthCm
5	3.9	1.7
6	3.4	1.4
7	3.4	1.5
8	2.9	1.4
9	3.1	1.5

```
In [ ]: #data formatting
```

```
In [ ]: df.dtypes
```

```
Out[ ]: Id                int64
SepalLengthCm          float64
SepalWidthCm           float64
PetalLengthCm          float64
PetalWidthCm           float64
Species                object
dtype: object
```

```
In [ ]: #data normalization
from sklearn import preprocessing
```

```
In [ ]: df2= pd.DataFrame(df,columns=["SepalLengthCm","SepalWidthCm","PetalLengthCm",
```

```
In [ ]: df2
```

```
Out[ ]:
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

```
In [ ]: x=df2[df2.columns].values.astype(float)
```

```
In [ ]: min_max_scalar=preprocessing.MinMaxScaler()
```

```
In [ ]: x_scaled = min_max_scalar.fit_transform(x)
```

```
In [ ]: df_normalized = pd.DataFrame(x_scaled)
```

```
In [ ]: df_normalized
```



```
Out[ ]:
```

	0	1	2	3
0	0.222222	0.625000	0.067797	0.041667
1	0.166667	0.416667	0.067797	0.041667
2	0.111111	0.500000	0.050847	0.041667
3	0.083333	0.458333	0.084746	0.041667
4	0.194444	0.666667	0.067797	0.041667
...	...	...	...	...
145	0.666667	0.416667	0.711864	0.916667
146	0.555556	0.208333	0.677966	0.750000
147	0.611111	0.416667	0.711864	0.791667
148	0.527778	0.583333	0.745763	0.916667
149	0.444444	0.416667	0.694915	0.708333

150 rows × 4 columns

```
In [ ]: df2['Species']=df['Species']
```

```
In [ ]: df2['Species'].unique()
```

```
Out[ ]: array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

```
In [ ]: label_encoder=preprocessing.LabelEncoder()
```

```
In [ ]: df2['Species']=label_encoder.fit_transform(df2['Species'])
```

```
In [ ]: df2['Species'].unique()
```

```
Out[ ]: array([0, 1, 2])
```

```
In [ ]: df2['Species'].dtype
```

```
Out[ ]: dtype('int64')
```

```
In [ ]: df2
```

```
Out[ ]:
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	2
146	6.3	2.5	5.0	1.9	2
147	6.5	3.0	5.2	2.0	2
148	6.2	3.4	5.4	2.3	2
149	5.9	3.0	5.1	1.8	2

150 rows × 5 columns

```
In [ ]: features_df=df2.drop(columns=['Species'])
```

```
In [ ]: enc = preprocessing.OneHotEncoder()
```

```
In [ ]: enc_df=pd.DataFrame(enc.fit_transform(df2[['Species']]).toarray())
```

```
In [ ]: df_encode = features_df.join(enc_df)
```

```
In [ ]: df_encode
```

```
Out[ ]:
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	0	1
0	5.1	3.5	1.4	0.2	1.0	0.0
1	4.9	3.0	1.4	0.2	1.0	0.0
2	4.7	3.2	1.3	0.2	1.0	0.0
3	4.6	3.1	1.5	0.2	1.0	0.0
4	5.0	3.6	1.4	0.2	1.0	0.0
...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	0.0	0.0
146	6.3	2.5	5.0	1.9	0.0	0.0
147	6.5	3.0	5.2	2.0	0.0	0.0
148	6.2	3.4	5.4	2.3	0.0	0.0
149	5.9	3.0	5.1	1.8	0.0	0.0

150 rows × 7 columns

```
In [ ]: df_encode.rename(columns={0:'Iris-Setosa',1:'Iris-Versicolor',2:'Iris-Verginica'})
```

```
In [ ]: df_encode
```

```
Out[ ]:
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Iris-Setosa	Versicolor
0	5.1	3.5	1.4	0.2	1.0	0.0
1	4.9	3.0	1.4	0.2	1.0	0.0
2	4.7	3.2	1.3	0.2	1.0	0.0
3	4.6	3.1	1.5	0.2	1.0	0.0
4	5.0	3.6	1.4	0.2	1.0	0.0
...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	0.0	1.0
146	6.3	2.5	5.0	1.9	0.0	1.0
147	6.5	3.0	5.2	2.0	0.0	1.0
148	6.2	3.4	5.4	2.3	0.0	1.0
149	5.9	3.0	5.1	1.8	0.0	1.0

150 rows × 7 columns

```
In [ ]: one_hot_df = pd.get_dummies(df2,prefix="Species",columns=['Species'],drop_first=True)
```

```
In [ ]: one_hot_df
```

```
Out[ ]:
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species_0
0	5.1	3.5	1.4	0.2	True
1	4.9	3.0	1.4	0.2	True
2	4.7	3.2	1.3	0.2	True
3	4.6	3.1	1.5	0.2	True
4	5.0	3.6	1.4	0.2	True
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	False
146	6.3	2.5	5.0	1.9	False
147	6.5	3.0	5.2	2.0	False
148	6.2	3.4	5.4	2.3	False
149	5.9	3.0	5.1	1.8	False

150 rows × 7 columns

```
In [ ]: col = ['Species_0', 'Species_1', 'Species_2']
one_hot_df[col] = one_hot_df[col].astype(int)
```

```
In [ ]: one_hot_df
```

```
Out[ ]:
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species_0
0	5.1	3.5	1.4	0.2	1
1	4.9	3.0	1.4	0.2	1
2	4.7	3.2	1.3	0.2	1
3	4.6	3.1	1.5	0.2	1
4	5.0	3.6	1.4	0.2	1
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	0
146	6.3	2.5	5.0	1.9	0
147	6.5	3.0	5.2	2.0	0
148	6.2	3.4	5.4	2.3	0
149	5.9	3.0	5.1	1.8	0

150 rows × 7 columns